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17 Transforming the Evaluation: The Crucial Role of Natural Language Processing in Intelligent Automation System

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Abstract

The story of NLP blending with intelligent automation systems (IAS) is an interesting chapter in technological history. Emphasis on this convoluted relationship between these two domains, we anticipate that by exploring the history of their interaction during a phase of rapid growth in NLP, to provide perspective into how developments within NLP have impacted IASs and vice versa. This chapter traces the change and does an analysis of the role that intelligent automation systems (IAS) in natural language processing have played in many sectors over the years, as well as a historical association linking these two fields. It considers the earliest IAS efforts and major strides in NLP such as phrase parsing and language modeling. The study goes on to cross-examine examination points showing the significant overlap of NLP that facilitates natural language conversation automation. This chapter also explores the embedded advantages of natural language processing (NLP) in intelligent automation and how utilizing NLP improves system skills in a way that will certainly streamline automation processes and help circulate human-machine communications consistently. By guiding the readers through a rapidly changing landscape of NLP-driven Intelligent Automation, it accomplishes its objective - to equip practitioners and stakeholders with an in-depth understanding of what is already out there, where things are heading next, who has solved (or not) which riddle before them; along offering possibilities they may want to consider and associated challenges ahead.

Keywords**:** Natural language processing, intelligent automation systems, sentiment analysis

17.1 Introduction

The aim of the foundational work in natural language processing (NLP) was the creation of algorithms for tasks like sentence parsing, part-of-speech identification, and language model generation that would enable computers to understand and interpret human language. These developments did lay the foundation to build intelligent automated systems that were able to automate tasks involving natural language interactions like text processing, user queries, and extracting information from docu-

ments. This was because of sophisticated technologies in natural language processing. Intelligent automation and natural language processing (NLP) present a powerful combination that is enabling transformation in many sectors. Artificial intelligence and natural language processing (NLP) help machines understand, and interpret human language to undertake tasks that include repetitive operations automated using robotic process automation (RPA) by intelligent automation. Integrating NLP with IA: Natural language processing (NLP) may enhance intelligent automation by making it possible for systems to understand unstructured data including social media posts, emails, and customer reviews. This makes it possible for businesses to automate difficult tasks requiring conversations in natural language, such as sentiment analysis, content moderation, and customer service. Whether it is enhancing customer experience, simplifying workflows, or making businesses more productive (by driving increased accuracy, efficiency, and scalability across company activities), there are a good number of cases where these complementary technologies can come together to contribute to business disruption for almost any industry sector. - The two dynamic forces of NLP and intelligent automation have the potential not only to transform many of the fields they touch but also to provide new vistas for innovation. - Plan of study, the argument is laid out as follows: - Introduction - To see how NLP can impact Intelligent Automation - An overview of Current Applications – “How Intelligent Automation Overview of Future Develop and Opportunities.” – Conclusion.

17.1.1 The Background: An Overview of NLP and Intelligent Automation History

Below, Figure 17.1 shows the development stages of natural language processing. The work of NLP as a field is to understand human language and to make it possible for computers to interpret and analyze human grammar. It involves creating methodologies and algorithms to help with the generation of language models, part-of-science tagging of words, and parsing or analyzing sentences. These early attempts at NLP laid the groundwork for later, successive iterations of NLP that helped make it possible for robots to produce and understand language better. Intelligent Automation, however, has to do with the fusion of automation

technologies, such as artificial intelligence (AI) and robotic process automation (RPA), for the purposes of optimizing and autonomizing various processes and workloads [1]. Technology was expected to use the first versions of intelligent automation systems to increase the efficiency and results of machine-driven repetitive tasks that people were previously doing.

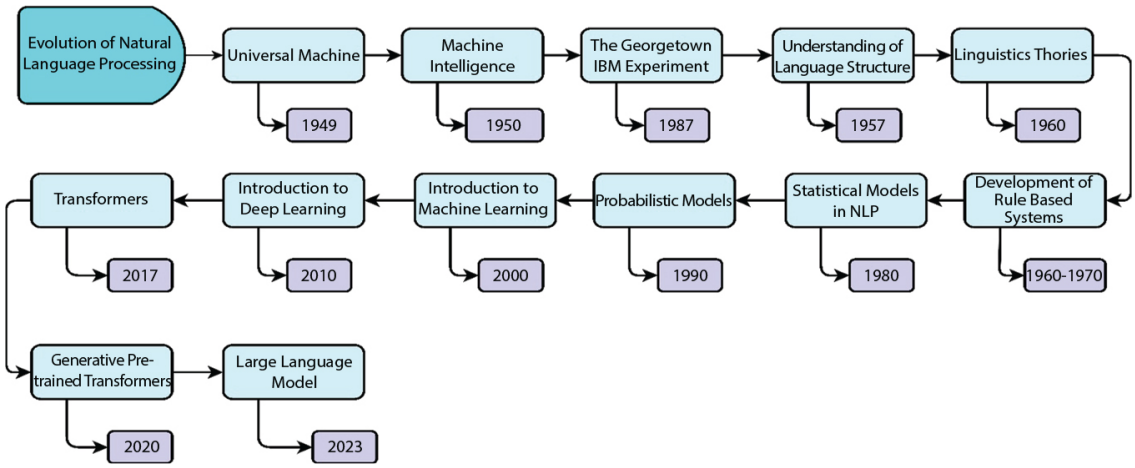


Figure 17.1 Development stages of natural language processing.

The key takeaway from this part is to provide a fundamental concept to science and later on research that will explore the coalescence and communication of the two fields-NLP and Intelligent Automation. Such as, how from its first year of existence NLP and Intelligent Automation have been weighing in on each other's achievements from the start. One way of knowing would be to examine its turning points and its inventive milestones.

17.1.2 Early Endeavors in Intelligent Automation Systems

The development of intelligent automation systems has been significantly influenced by natural language processing or NLP. Significant contributions of NLP are exemplified by milestones ranging from universal machines to contemporary improvements. Robust methods for natural language processing tasks and automation were made possible by early advances in rule-based systems and statistical language processing. The rising synergy between NLP and automation was highlighted concurrently by significant milestones in the development of intelligent automation systems, such as the introduction of robotic process automation (RPA) and the integration of AI technology. Because robots are enabled to com-

prehend and analyze human language by NLP, it has been made easier for cognitive functions to be automated, which has revolutionized industries and increased productivity [2].

The development of intelligent automation systems has been significantly influenced by natural language processing (NLP) in several ways:

- How the evolution of intelligent automation systems has been significantly influenced by natural language processing or NLP.
- Human-Machine Interaction: It is thus possible for machines to understand and interpret human language using natural language processing, making them interact more intuitively with automated systems [3]. It has been achieved by the latter in constructing chatbots, and voice-enabled virtual assistants like Google Assistant, Amazon, and Alexa are capable of effectively receiving commands and promptings through human utterances.
- Data extraction and understanding: NLP helps to extract structured data from unstructured sources such as emails, web pages, or text documents. It performs automated processing and comprehension of masses of texts thus making smart systems able to make inferences, detect patterns, and valid judgments based on knowledge [3].
- Automation: NLP is very important for tasks that have to do with text-based inputs or outputs. They can be used to perform various activities that involve texts more efficiently like email classification, document summarization, sentiment analysis, and text generation.
- Knowledge Representation and Reasoning: The knowledge obtained from natural language materials is represented in NLP format, which allows reasoning on it. Included among these are methods for enabling knowledge-based reasoning and decision-making systems with intelligent automation to build and utilize structured knowledge bases like named-entity recognition, relation extraction, and knowledge graph production [4].
- Language Translation and Localization: Intelligent automation systems using NLP have been used in language translation and localization processes that involve handling multilingual data. Such applications are highly important for multinational firms dealing with different languages or regions as they enable the automated process-

ing of data in multiple languages thereby facilitating inter-language communication and automation [5].

- Contextual Understanding: With advanced natural language processing (NLP) techniques such as contextual embeddings and transformers, there has been a dramatic increase in the number of words understood by an intelligent automation system.

Generally, the incorporation of natural language processing (NLP) skills has greatly contributed to the development of intelligent automation systems. Comprehension skills have increased resulting from which human language can be fully grasped. The unstructured information can be obtained from them, text-based tasks can be automated hence making it more organic, and intuitive human-machine interactions can also be achieved [6].

17.1.3 Understanding the Essence: NLP's Contribution to Intelligent Automation

Natural language processing (NLP) has a very vital role in intelligent automation. The study highlights the role of natural language processing (NLP) in improving the capabilities of automated systems, as illustrated in Figure 17.2.

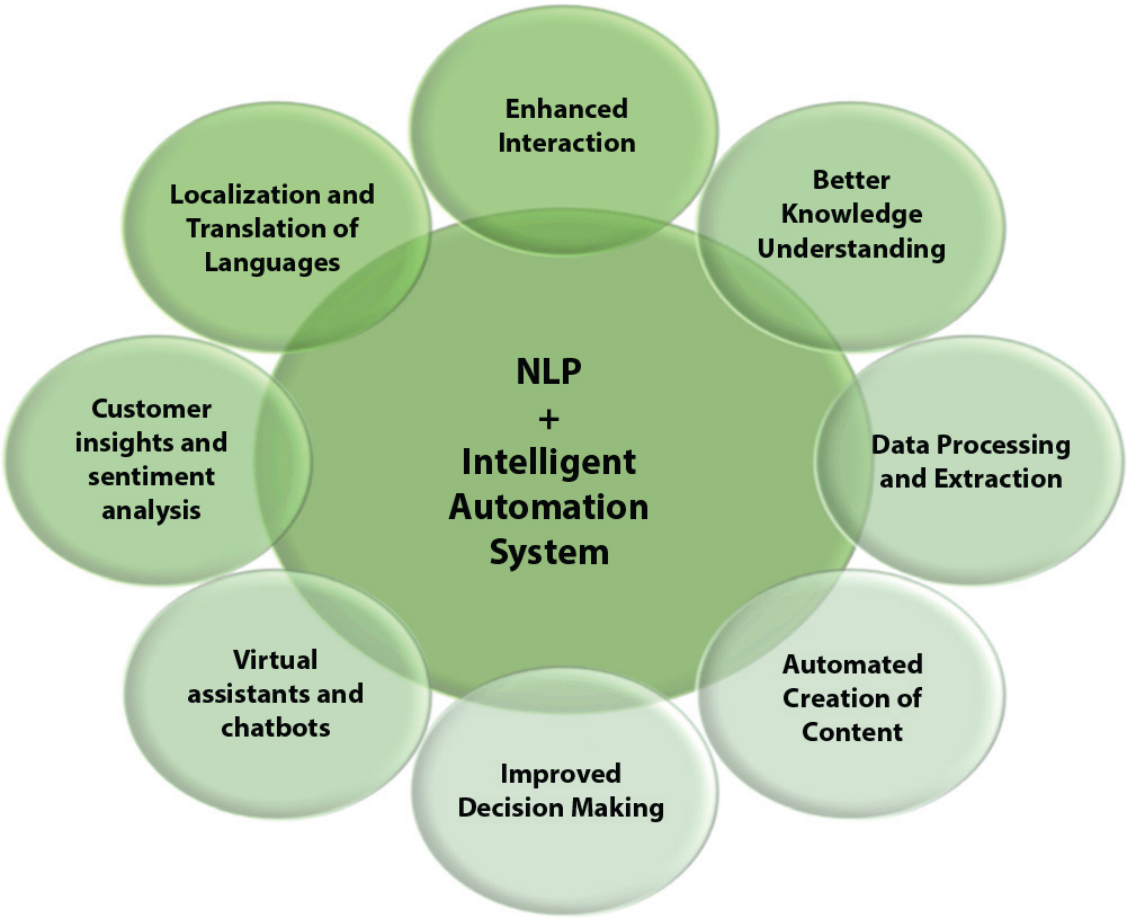


Figure 17.2 Natural language processing contribution to an intelligent automation system.

- Improved Communication: Natural language processing enables robots to comprehend, decode, and respond to human speech, enhancing user interaction and user experience [5, 7].
- Deeper Understanding of Knowledge: NLP improves text and language comprehension in machines. In these situations, automated systems' ability to handle massive volumes of data helps with information extraction, sentiment analysis, and summarization [7].
- Data Processing and Extraction: NLP helps extract relevant information from unstructured data sources, such as papers, emails, and social media. Content analysis, document classification, and data input automation can all benefit from this functionality [8].
- Automated Creation of Content: Content, summaries, and reports can be generated by NLP algorithms. Time and resources can be saved through this automation process [8].
- Improved Decision-making: Improved decision-making procedures are achieved through NLP, which makes intelligent automation possible. They can understand and analyze any natural language input to come up with such different judgments like responding to simple consumer inquiries or advising on complicated corporate strategies [9].

- Virtual Assistants and Chatbots: NLP is used by chatbots and virtual assistants alike. These systems use natural language understanding to do a variety of tasks and reply to queries, making the user experience more interactive and pleasant [10].
- Enhancing Workflow and Automating Processes: By integrating NLP, workflow automation may be improved and corporate operations can be optimized. Automated systems that use language analysis and interpretation to initiate and manage tasks can increase overall accuracy and performance [10].
- Customer Insights and Sentiment Analysis: NLP is crucial for analyzing customer sentiment from social media, reviews, and comments. Decisions based on facts, products, and services can be made with the use of this data. It might result in more satisfied customers [11].
- Localization and Translation of Languages: Automatic language translation is made easier by NLP, creating global commercial potential. This is especially advantageous for companies with diverse customer and global operations.

In summary, natural language processing plays an important role in the development of Intelligent Automation by allowing machines to comprehend and interpret human language, which increases the adaptability, interactivity, and task-handling capacity of automated systems [12].

17.1.4 Reviewed Studies

Table 17.1 describes some reviewed studies related to the particular area. This table shows the specific technique that is used by authors, the application area where that technique was applied, followed by advantages and limitations.

Table 17.1 Reviewed studies.

Ref.	Techniques	Application	Advantages	Limitation
[1]	NLP, machine learning, clustering, and topic modeling	Sentiment analysis in social media, document summarization	Provides consistent analysis and processing of text without the variability introduced by human workers. Reduces the need for manual processing, thereby lowering operational costs.	Training and running advanced NLP models can be resource-intensive, requiring significant computational power and memory.
[7]	Robotic process automation, AI, machine learning, NLP	Finance and accounting, healthcare	RPA combined with AI significantly speeds up processes by handling large volumes of tasks with high accuracy.	High initial costs for setting up AI and RPA systems, including software, infrastructure, and training.

Ref.	Techniques	Application	Advantages	Limitation
[9]	NLP, machine learning, Chatbots, and virtual assistants	Recruitment and selection, employee onboarding, employee engagement	Speeds up recruitment and hiring processes. Automates repetitive tasks, freeing HR staff for strategic activities Handles large volumes of data and applications efficiently. Supports growing organizations without a proportional increase	Handling sensitive employee data raises privacy and security issues. Initial setup and integration of AI systems can be expensive
[10]	Sentiment analysis (text using word embeddings (Word2Vec, GloVe) and contextual embeddings (BERT, GPT)).	Language translation, content generation	Advanced NLP models achieve high accuracy in understanding and processing human language. Reduces the need for manual labor	Some advanced NLP models, especially deep learning models, act as “black boxes,” making it difficult to understand how they make decisions. This

Ref.	Techniques	Application	Advantages	Limitation
			in text processing, leading to cost savings.	lack of interpretability can be a significant drawback in critical applications where understanding the decision-making process is crucial.
[18]	Feature extraction: frequency-inverse document frequency (TF-IDF), and word embeddings (Word2Vec, GloVe). Supervised learning algorithms, long short-term memory (LSTM) networks, and transformers (e.g., BERT)	Customer feedback analysis, market research	Provides valuable insights into human emotions and sentiments, aiding in better decision-making Enables real-time monitoring of sentiment and emotions, which is crucial for timely interventions and responses.	Sentiment and emotion detection models may struggle with understanding context, sarcasm, and irony, leading to inaccurate results.

Ref.	Techniques	Application	Advantages	Limitation
[19]	Linguistic knowledge acquisition, information extraction, text summarization	Document management, information retrieval	Automates the process of extracting and summarizing information, saving time and resources.	Performance depends on the quality and coverage of the linguistic knowledge base and the patterns used for extraction.
[13]	LLMs	Intelligent assistant	LLMs are trained on large amounts of data, difficult to understand	Understanding and adapting individual user preferences, needs, and contexts.
[14]	NLP-based POP-ON	Business process	Predicts failure or variations of business process	Difficult to understand
[20]	Lexicon, topic modeling, and word embeddings	Smart education	Findings about minority representation in textbooks	Domain-specific lexicons not available
[25]	IoT	Smart healthcare domain	Deep analysis of specific security and privacy features	Specific domain

Ref.	Techniques	Application	Advantages	Limitation
[28]	Natural language processing	Smart city	NLP applications in healthcare, industries, business, and legal management	Open challenges like multilingual conversations, linguistic variations, and part-of-speech tagging
[31]	Natural language processing	Medical domain	Privacy preservation	Limited to medical and technological domain
[32]	Machine learning algorithms, speech recognition technology	ELT pedagogy	Personalized learning experiences	Data privacy and security
[35]	NLP and AI techniques	Education	Adaptive and personalized teaching process	Fail to reflect the most recent development
[37]	IoT, NLP, and Cloud	Home automation	Control device through voice	Specific domain
[39]	NLP and ML: LR, NB, and K-SVM	Smart society	Analyzed social work activities in public care	Small dataset size

17.2 Natural Language Processing (NLP) as the Foundation of Intelligent Automation

Natural language processing (NLP), augmented by the power of language, is the bedrock of intelligent automation and is changing how machines are interacted with by people as well as how data is governed. NLP improves the effectiveness and efficiency of automated systems, as they can easily understand, interpret, and respond to human language. NLP also underpins activities such as sentiment analysis, information extraction, and content summarization where raw textual data is converted into actionable insights. Natural language processing (NLP) empowers chatbots and virtual assistants to have natural, conversational dialogs with users. NLP's capability to extract information from unstructured sources facilitates a wide range of processes from intelligent decision-making to automating document categorization [7, 13].

17.2.1 Improving Efficiency: How NLP Can Help Streamline Automation Processes

Natural language processing (NLP) has been a breakthrough in automating business operations for better productivity. By training machines to understand and work with human language, NLP makes easy the way users interact (from an automated system standpoint). This is particularly relevant in data-centric roles as natural language processing (NLP) can be used to uncover valuable insights from unstructured sources like emails, papers, and social media. The ability to understand natural language can help identify data with more speed and accuracy leading to faster results. Making sense of unstructured data by NLP helps in several operations right from helping in intelligent decision-making and document classification automation to name a few.

Natural language processing (NLP), for example, allows automated systems to read and process text as well as determine meaning more rapidly. For example, NLP increases the speed and accuracy of responses in customer-facing apps such as chatbots resulting in a more efficient and satisfying user experience [14].

17.2.2 The Role of Natural Language Processing (NLP) in Enabling Seamless Communication Between Humans and Machines

Natural language processing or NLP in short is a very important asset for bridging the gap between people and machines, as well as providing natural interaction communication. This was one of the key features that allowed it to break through the traditional walls between systems for users and RPA. Natural Language Processing: Converses in human languages, making interactions more natural for humans and robots. Natural Language Processing Interface: Chatbots, virtual assistants, and other NLP-driven interfaces. This means the user can talk to a machine like a human when a chatbot/virtual assistant is developed (the progress of any other bots). This improves the experience of the software being used while allowing more and more people to make use of new technology. Humans use their abilities to understand context, intent, and minor differences in human expression by parsing the nuances of language while robots can do this with tools that process textual-based techniques known as NLP [15].

17.3 Exploring Current Applications

NLP plays an important role in various application areas that are given below.

17.3.1 Revolution in Customer Service: Chatbots and Virtual Assistants

Natural language processing is very important in reshaping client services with virtual assistants and chatbots. It enhances interactions with clients by offering personalized affairs based on particular requirements. Virtual assistants and chatbots offer proper responses based on language patterns and context which results in a high level of engagement and satisfaction. NLP also allows for the continuous development and learning of the information. Machine learning algorithms are utilized to provide a higher level of accuracy, and effectiveness by reacting to the user inquiries and comments. Furthermore, using NLP resource utilization can

be maximized by executing the operations in repetitive processes and reducing the human burden to concentrate on this thing. In summary, NLP provides users with efficient, scalable, and personalized experiences that also increase customer satisfaction and loyalty in a digital world [16].

17.3.2 Text Analytics and Sentiment Analysis

Tokenization and the elimination of stop words are the fundamental steps in preprocessing when dealing with text data. Syntactic analysis can be supplemented by named entity Recognition (NER), which helps to find names and words like locations and dates. POS tagging is the procedure of labeling words to their grammatical category for a clearer meaning of location. The tone of a text is established by attitude analysis [40, 41]. LDA or NFM are examples of topic modeling techniques that categorize and summarize collections of texts. While using identical kinds of text, document summaries and text classification help automate categorization. Relation extraction defines the relation and the improved models such as BERT add more contexts to it. Analyzing customer feedback and social media are some of the activities well done by intelligent automation through NLP that enhance decision-making and customer satisfaction [17]. For example, sentiment analysis for social media monitoring: An online retailer needs others' perceptions about the company and its brand on social networks. This process of analyzing opinions focuses on categorizing the feedback more profoundly by distinguishing between positive, negative, and neutral feedback on the base of the analysis of the tweets, comments, and posts of the users from Facebook or Instagram. The system notifies the social media staff when negative opinions are rising so that they can attend to them promptly. Perceived advantages also help to determine the strategies for placing advertisements and increasing consumers' commitment. In conclusion, it may be stated that when NLP is integrated with text analytics and SA, it becomes an effective tool for gaining insights from large text data in the context of enterprises. These applications will continue to be important as automation systems progress in the future as a way of increasing efficiency and effectiveness in making decisions [18].

17.3.3 Information Management by Extracting and Summarizing Documents

Effective knowledge management requires NLP, in particular for tasks such as information extraction and document summarization. Document summarizing helps in reducing long papers to short summaries that still contain important information and remove fluff. In an enterprise context, NLP-based document summarizing could help convert lengthy reports or research papers to easily understandable insights for decision-makers therefore saving time. Information extraction on the other hand is about extracting relevant pieces of information from unstructured textual sources like emails, documents, or even customer reviews. Using NLP techniques such as named entity recognition (NER) and Relation Extraction among other things, organizations can automatically extract key data points required to drive insights and analytics efficiently. NLP helps intelligent automation systems not just in improving management processes but also in ensuring the right pertinent insights are available across domains/verticals.

17.3.4 Role of Adaptive Learning in the Training System

Using NLP, you can easily employ agents like chatbots or virtual assistants to create interactive learning experiences. Such resources can deepen understanding by offering immediate feedback and responses to queries, as well as support in context-aware learning. Intelligent automation systems have integrated the application of natural language processing (NLP) in instructional and training systems. This allows learners to progress through the content at their own pace, focus on topics that catch their interest, and look for help. It increases student performance across all types of learning environments and academic domains. In IAS, NLP is becoming more crucial for learning and course recommendation in personalized e-learning and training applications. They offer a more personalized learning environment as such systems can address the needs and demands of each learner. The functioning of this system is as follows; NLP algorithms are applied to different aspects of students' interactions with content including learning modality, understanding level, learning rate, and question-solving methodologies. Thus, by examining the language

that is used by the learners, NLP algorithms can identify some patterns and trends in their learning behavior [20]. The system does this by adapting its learning resources and the tasks given to each student for learning and practice. For instance, if a pupil fails to understand a particular concept, then the system may proceed to explain to them for the sake of the pupil or refer them to additional notes that have been designed with the pupil in mind. For example, for those students who have demonstrated proficiency in a specific topic, it can also assist in motivating them into challenging activities or classes.

Furthermore, ease of use makes it possible to incorporate the chatbot or virtual assistant in developing an engaging learning experience. Such resources may enhance learning in general, since they provide instant feedback, address questions, and/or provide guidance in the course of the learning process. Intelligent automation systems have emerged as the core elements for enhancing education as well as training through the incorporation of NLP in adaptive learning solutions. It enables learners to move forward at their rate complete lessons that they find fascinating and get a lot of support in areas they find terrible. This approach improves learning results both in terms of general context and specific subjects.

17.4 Future Directions for NLP in an Automated Environment

Being an advancing technology, NLP is positioned to transform the sphere of intelligent automation as the development of that field progresses. NLP will be more critical in the future as it supports leveraging automatic processes, understanding language context, and decision-making processes. It also paves the way for future developments of multilingual facets, advanced language models, and even more natural interactions between man and machine [21].

17.4.1 NLP with Intelligent Automation

In natural language processing (NLP), context sensitivity means imparting the ability to machines to “manage” the uncertainties, challenges, and

nuances of language inherent in natural communication. This capacity involves extracting people's purpose and feelings as well as contexts within talks in addition to keywords. For NLP, there is a challenge of distinguishing one figure of speech from another including metaphors, sarcasm, and cultural references to enhance the understanding offered to users. CUAMR aims at trying to take the sophistication of NLP systems to the next level to allow for an enhanced level of flexible and powerful communicative patterns to hold and at the same time try to enhance the levels of accuracy in human-machine interfaces. An enduring issue that both computer science and natural language processing face remains the issue of how to close the gap between the idea of a computer understanding and the complexity of the actuality of language representation [22, 23].

17.4.2 Integrated Multimodal

In natural language processing (NLP) context-awareness is the process of training machines to allow for vagaries, convolutions, and nuance associated with languages. This capacity involves the ability of systems to grasp not just literal meanings of words and phrases as used in the talks but also the mood and intents behind them and any contextual pointers. The challenge to NLP is to accentuate distinct aspects such as metaphors, sarcasm, and cultural references for a better understanding of the user. Figure 17.3 describes the contextual understanding of natural language processing which aims at increasing the complexity of current natural language processing (NLP) systems in a way that allows them to deal with various approaches to communication, making human interaction processes more accurate [24].

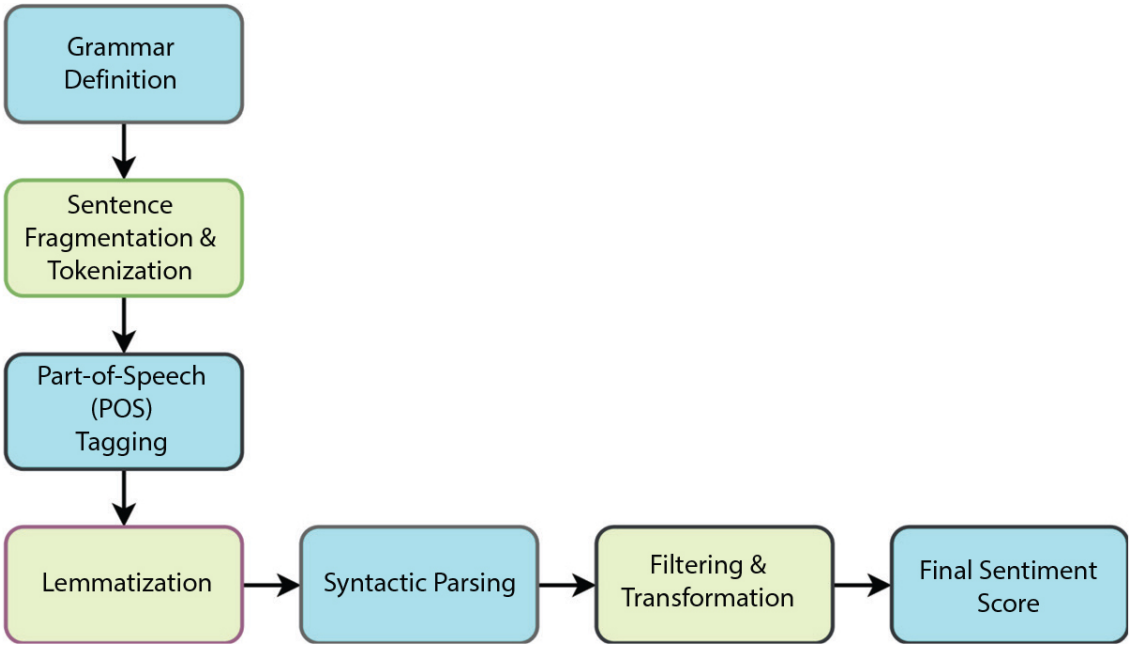


Figure 17.3 Contextual understanding of NLP.

17.4.3 Ethical Imperatives: Ensuring Responsible AI in NLP-Driven Automation

When it comes to automation through natural language processing, the issue of ethics is paramount, which underlines the need to maintain ethical form in creating artificial intelligence. Since NLP is at the core of intelligent process automation, ethical questions are paramount to ensure fairness, neutrality, and accountability of solutions. In this context, issues on algorithmic bias, data privacy, and appropriate utilization of NLP technology are forced ethical considerations. There is a need to ensure that NLP-based automatic systems operate on ethical considerations not to have unforeseen consequences: being biased or abused. To advance the idea of responsible and ethical AI practice and strive for the ethical integration of AI in society, it is important to thus propose rules and regulations, make AI decisions transparent, and several voices inclusive in the creation of the NLP-driven automation [25].

17.4.4 Zero-Shot Learning: Leveraging Little Data to Adapt to New Tasks

A potential is that in zero-shot learning, models may be trained to learn problems that it has never encountered before unlike in normal machine learning where it requires a huge amount of labeled data. This is made possible through context mechanisms and semantic relations that allow

the model to reason from patterns and relations even where there are no specific sample instances to train it. This is advantageous in cases when it is not possible to obtain labeled data for each activity of interest. Zero-shot learning is a significant advance in artificial intelligence that contributes to the principle of generalization and versatility. Thus, it may have a place in various areas, including natural language processing and picture recognition [26].

17.4.5 Adaptation Based on Domain Area

Domain specificity in natural language processing (NLP) works by changing the models to fit the specific needs of certain fields. This procedure aims at trying to fine-tune NLP models for specific disciplines, sectors, or activities in light of the domain specificity of language use. With the help of fine-tuning existing models based on different domain datasets, the experts may make NLP solutions more effective and relevant to particular fields, such as healthcare, finances, or document handling. When the NLP models are trained to understand the language, terminologies, and contextual requirements specific to a particular subject or field, it makes the learning process more reliable and will give better results in terms of that specific subject or field. In more specific contexts, different and focused customization may enable NLP to correct fine peculiarities of special languages, and that means many splendid application options will be revealed [27].

17.4.6 Explainable AI: Demystifying NLP's Decision-Making Processes

Domain adaptation, still in the context of NLP, means tailoring models to fit the requirements of different uses. This procedure aims at tuning NLP models for specific disciplines, sectors, or activities while acknowledging the differences in language usage across domains. With the help of the refinement of pre-existing models with the help of domain-adapted datasets, such professionals might enhance the effectiveness and the relevance of the uses of NLP in different fields like healthcare, the financial sphere, or legal documentation. It also provides certainty that a range of domain-specific terminologies, phrases, and contextual features are cor-

rectly and semantically interpreted by NLP models [28]. In this way, by increasing the focus on specific industries, NLP can be made to adapt to the requirements of specific industries and address the many small, specific linguistic peculiarities; new opportunities for application will open up.

17.5 Challenges and Opportunities

NLP combined with intelligent automation systems has potential benefits and threats in different domains.

- Data Quality: Data models depend on the quality of training data. One of the persistent problems remains the true and timely data acquisition for training [3].
- Variability in Language: Languages also have variations in the grammatical structure, syntactic and semantics. Dealing with such variation is a challenge in constructing efficient NLP models that can capture different language patterns [29].
- Ambiguity and Context: Reducing ambiguity and account for context in natural language is still a problem for NLP systems, especially in cases of sentimental analysis or processing of sarcasm and jokes [30].
- Ethical and Bias Concerns: Some of these models can accidentally perpetuate biases that are inherent in the data set on which they are trained and may lead to unfair outputs or actions or outcomes that are ethically questionable.

17.5.1 Addressing Privacy and Security in NLP-Driven Automation

NLP-automation is badly affected by various security and privacy issues. Below, some suggestions are provided to handle these issues [31].

- Proper Authorization and User Authentication: In an automated environment, a proper security procedure must be followed and a strong security and authentication mechanism should be applied so that only authorized parties can access the important information. Person privacy should be maintained by using PII (personally identifiable infor-

mation. This makes the user's identity unidentifiable. It also provides security while automation and data analysis take place.

- Data Privacy: Some privacy mechanisms should be applied while data is in transit or at rest by encrypting the important data. This will ensure that data will be in a secure state even if the third party steals the information.
- Data Handling: Gathering important data and ignoring the redundant or useless data result in an optimal solution. When the NLP model is trained, it is very important to select the dataset very carefully that fulfills the requirements and follows the proper ethical principles. Techniques like federated learning and differential privacy can be used to train the NLP model securely.
- Regular Security Audits: Proper security audits should be there to make sure that the proper standards are followed and a strong monitoring system should be there to address the tragedies that happen in a system.
- Risk Management: It is necessary to manage the risks that are associated with third parties and vendors. Users should properly inspect that third is using proper security standards and following some ethical guidelines before using their services.
- Infrastructure Management: By implementing best practices in the NLP automation system, proper infrastructure can be managed. Time to time software updates, software patches, and access control mechanisms should be applied in the automated environment.
- Monitoring System: By checking proper resources and service requirements, the system can be updated for further process. Therefore, proper monitoring should be there to inspect the elements within the automated environment.

17.5.2 Filling the Gap: Promoting Inclusivity and Accessibility

Some accessibility and inclusion gaps should be addressed as a priority while users are deploying and developing the NLP-based solutions.

- **Globally Diverse:** The NLP-based automation system should support multi-languages. Therefore, it is easy for the users to understand at a

global level. People from different countries or regions can understand this irrespective of their geographical background [32].

- **Cultural Diversions:** The NLP-based automation system should always consider cultural diversion. Automated communication *a priori* in the system should be avoided and nuances should always be kept in mind while developing the automated system [33].
- **User Feedback:** For identifying the problems and solutions to those problems, it is important to take the proper feedback and improve based on that. For taking the proper feedback, people from diverse backgrounds and minority languages should be considered. Based on the acquired feedback, the proper improvement should be there [34].
- **Training and Education:** Assist members of underrepresented populations to understand how to implement NLP technology-based automation by giving them training and informational resources. This can be done by offering them tailored instructions in the form of tutorials and workshops and supporting them through forums. It is crucial to train flexible NLP models using diverse data that reflects the potential audience to design accurate language models. To fine-tune the machine for processing language inclusively, data from a vast number of languages, dialects, and speech must be included [35, 36].
- **Social and Ethical Considerations:** It focuses on how bias, discrimination, and privacy are affected by the use of NLP automation and how it will result in the output. It provides ethical consideration to the users to make fair use of it to increase efficiency and effectiveness [36, 37].

17.6 Developing Talent: Cultivating Natural Language Processing Masters of Tomorrow

Future breakthroughs and innovation in the NLP domain will be determined by the next generation of professionals; therefore, there should always be fast development in the fast-growing field. It is highly important to create structures for learning activities that can cover more complex topics in NLP [38, 39]. These programs may contain online resources, seminars, and courses to cater to all levels of students. NLP can be used in practical NLP projects that make it easy for the students to gain real knowledge and practical experience. It also handles tasks including text

categorization, sentiment analysis, speech recognition, and language translation [38, 40, 41].

17.7 Conclusion and Future Scope

In summary, natural language processing is the most innovative and trending technique that is very beneficial in developing the automated system. It demonstrates how the NLP has evolved and transformed from the human-machine interface and still evolving in several industries. Efficient transformer-based systems are becoming very popular nowadays and also various deep learning mechanisms are highly contributing to this to enhance the efficiency and efficacy of the system. It is playing a crucial role in areas ranging from information management to customer services. Natural language processing in collaboration with intelligent automation systems is developing very fast and it is playing a great role with respect to the development of intelligent systems. The development of NLP will be shaped by unsupervised learning, multimodal NLP, and ethical issues, which results in a more immersive, responsible, and adaptive human-machine interaction model.

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